DI2C IN
DI2C OUT
U2 PCA9615DP
J1 Conn_01x06
J4 Conn_01x06
IC4 STM32L433CCU6
IC2 TB67S249FTG,EL
J2 5077CR-16SMC2-BK-TR
U3 USBLC6-2P6
J5 SENS I2C (3V3/GND/SCL/SDA)
J6 POWER 12-24V / >=10A (1=VIN,2=PGND)
Q1 IRF4905
F1 T10A/60VDC
D6 SMCJ26A
D7 BZT52C12 (12V)
U5 LMR36510ADDAR
U6 TPS2116DRLR
U7 AP2112K-3.3TRG1
U8 TMUX1511PW
L1 Ferrite 600R@100MHz / >=200mA
TH1 Thermistor_NTC
SW1 SW_MEC_5G
SW2 SW_MEC_5G
SW3 SW_MEC_5G
SW5 SW_MEC_5G
NT1 PGND-GND STAR
NT2 PGND-GND STAR
MCU STATUS (HIGH = ON)
COMM STATUS (HIGH = ON)
RUN / ACTIVITY (MCU, HIGH = ON)
FAULT (MCU, HIGH = ON)
MOTOR A+/A-/B+/B-
5V BUCK / 400kHz / 1A
EXTERNAL FIRST / USB BACKUP
3.3V MCU SUPPLY
VM MEASUREMENT / PA0 ADC1_IN5
U1
U4

TERM: POPULATE AT BUS ENDS ONLY

USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9: GPIO VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_CR2.USV in firmware.
Internal D+ pull-up / termination. 90-ohm differential PCB pair.

AS5600 external sensor / 3.3V I2C
STM32 I2C1: PB6=SCL, PB7=SDA (AF4)
J5: 1=3V3, 2=GND, 3=SCL, 4=SDA
Fit one set of pull-ups on the bus.

BOOT0 uses existing boot switch/pulldown (PB8_BOOT net name is legacy).
Use HSI/MSI internal clock for USB standalone startup.

POWER INPUT 12-24V (J6) / reverse polarity + transient protection

STM32L433 IC4 <-> TB67S249 IC2 - stm32l433_tb67s249_connection.md

LO1/LO2/MO pull-ups: +3V3_D only (external digital 3.3V supply).
VM and +3V3_D supplied by power circuitry outside this specification.

AGC straps: R50=0; R52/R54/R56=100k; R51/R53/R55/R57=DNP.
CLIM1/BOOST/FLIM pull-ups must use DRV_VCC, never +3V3_D.

OSCM: R23=5.1k, C20=270pF; ~67kHz.
Place close to pin35. DRV_VCC: internal 5V, setting circuits only.

USB only: MCU on, motor VM off.
USB current before enumeration <=100mA total.
VM = ADC/4095 * VDDA * 19; wait >=10ms after power changes.
4.5A is driver absolute peak maximum; initial VREF limit ~1.97A/phase.
Route high-current PGND via NT1 only; keep BUCK_SW away from ADC.
AP2112 dissipation and motor/Q1 cooling require thermal validation.

VM capacitors: C15 bulk 220uF/63V; C16 at VMA; C17 at VMB; C18 10uF/50V.
Keep VM-PGND loops short; paired pins joined at IC on PCB.

VREF: R29=10k series, R30=9.1k to GND, C21=100nF.
3.3V DAC full scale -> VREF 1.57V -> IOUT 1.97A.
R30/C21 GND return near pins29/30, away from OUT/VM.

NT1: single PGND-GND copper connection directly beneath IC2 exposed pad.
EP/corner pads to GND, dense thermal vias; 4 layers / 2oz. Motor outputs: 5A routing.

Firmware: DAC1_OUT1 PA4 buffer enabled; reference VDDA/VREF+ (+3V3_A).
Buffered DAC may retain ~0.2V: ENABLE Low or disable DAC for zero current.
Startup: check VM -> set DAC -> set DMODE -> wait >=7.5us -> ENABLE High.
Reset: ENABLE/DMODE/AGC Low; evaluate initially with AGC0=AGC1=Low.

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